### 2.2 CoI Construction

Generating novel research ideas requires a profound comprehension of the respective research domain, coupled with a rigorous reasoning process. Previous endeavors (Lu et al., 2024; Baek et al., 2024) have sought to augment LLMs with relevant papers to facilitate the ideation process. However, these methods simply mix these papers into the prompt without effective organization. This scenario is akin to dropping an LLM at a chaotic intersection with no map in sight, leaving it uncertain about which path to take. To address this issue, we propose a Chain-of-Ideas agent framework.

As shown in Figure 2, a CoI, represented as $\{\mathrm{I}_{-M} \rightarrow \cdots \rightarrow \mathrm{I}_0 \rightarrow \cdots \rightarrow \mathrm{I}_N\}$, is a sequence consisting of $M+N+1$ ideas extracted from $M+N+1$ research papers respectively, where they together show the evolution progress within a given research field. Specifically, given an initial research topic, we prompt the LLM to generate multiple queries, $[q^1, \ldots, q^K]$, that reflect $K$ different perspectives of this topic. The prompt is given in Table 7 of Appendix. Unless otherwise specified, all prompts of our framework are presented in the Appendix tables. The $K$ queries are used to construct $K$ branches of CoI. This reduces the reliance on a single CoI that may be insufficient to capture the most significant development and direction. For each query $q^k$, we use it to retrieve a top-ranked paper, which we call anchor paper $\mathrm{P}_0^k$. In Figure 2, ToT (Yao et al., 2024) is an illustrative example of an anchor paper. An anchor paper serves as the foundation for constructing a CoI. Specifically, a CoI is constructed by extending from the corresponding anchor paper to related papers in both directions: forward, tracing the progression of ideas, and backward, tracing their origins.

In the forward direction, starting from $\mathrm{P}_0^k$, we identify subsequent papers that directly cite it by leveraging the Semantic Scholar API[2]. We use OpenAI's `text-embedding-3-large`[3] to rank these papers based on their cosine similarities to the concatenation of the initial research topic and the abstract of the anchor paper. Subsequently, we select the highest-ranked paper as $\mathrm{P}_1^k$ to extend the CoI in the forward direction (e.g. GoT in Figure 2). This process is repeated iteratively from $\mathrm{P}_i^k$ to $\mathrm{P}_{i+1}^k$, until either the length of the CoI reaches a preset value or the LLM finds that there is no valuable follow-up work (Table 8).

In the backward direction, starting from the anchor paper $\mathrm{P}_0^k$, we instruct an LLM to thoroughly review the full paper and to identify candidate references based on the following criteria: 1) references that $\mathrm{P}_0^k$ directly built upon, 2) references that serve as baselines in $\mathrm{P}_0^k$, and 3) references that tackle the same topic as $\mathrm{P}_0^k$. With those candidate references, we ask the LLM to determine the most relevant one to the anchor paper (Tables 9 and 10), denoted as $\mathrm{P}_{-1}^k$ (e.g. SC in Figure 2), to extend the CoI backward. This backward extension is also carried out iteratively from $\mathrm{P}_{-i}^k$ to $\mathrm{P}_{-(i+1)}^k$ to identify preceding papers (e.g. tracing backward from SC to CoT in Figure 2). It terminates when the length of CoI reaches a preset value or we encounter a milestone paper (defined as one with over 1,000 citations), indicating that the idea from the milestone paper could serve as a strong starting point for the CoI. Additionally, we instruct the LLM to terminate the search if no reference relevant to the original research topic is found (Table 8).

After we collect $K$ paper chains, denoted as $\{\mathrm{P}_{-M^k}^k \rightarrow \cdots \rightarrow \mathrm{P}_0^k \rightarrow \cdots \rightarrow \mathrm{P}_{N^k}^k\}_{k=1}^K$, we ask the LLM to extract ideas from these papers and inherit the progressive relation of the paper chains to form our CoIs $\{\mathrm{I}_{-M^k}^k \rightarrow \cdots \rightarrow \mathrm{I}_0^k \rightarrow \cdots \rightarrow \mathrm{I}_{N^k}^k\}_{k=1}^K$ (Tables 9 and 10). Then for each CoI, we ask the LLM to summarize the existing research trends by analyzing the evolution between any two adjacent ideas (Table 11). For example, the upper part of Figure 2 shows the evolution process from CoT to GoT step-by-step. Additionally, we extract experiment designs and the definition of key entities from these papers (Tables 9 and 10). The above information including CoIs and the derived knowledge will be used in the following idea generation and experiment design stages.

### 2.3 Idea Generation

In this section, we use the above-constructed CoIs and their developing trends to guide the generation of a novel idea. For each generated CoI, the first step is to predict possible future trends. As shown in the lower-left section of Figure 2, we prompt the LLM with the CoI, the developing trends of

[2] `https://www.semanticscholar.org/product/api`
[3] `https://openai.com/index/new-embedding-models-and-api-updates/`

existing works, and the key entities extracted from existing literature, as described in Sec. 2.2 (Tables 12 and 13). These entities comprise relevant datasets and potential baseline models, which are important to clarify the concepts mentioned in the existing literature. After obtaining the future trend, we continue to prompt the LLM to articulate its motivation, novelty, and methodology, finally consolidate the idea (Tables 14 and 15). Through this step-by-step manner, COI can produce a more detailed idea. Following the previous practice (Wang et al., 2023; Lu et al., 2024), we also use a novelty-check agent to evaluate candidate ideas. It retrieves relevant papers and prompts another LLM to assess the similarity between the generated idea and the retrieved papers (Table 16). Based on this assessment, our framework determines if another round of generation is necessary. Finally, we pairwisely compare the generated ideas from all CoI branches and select the one with the highest winning rate as the final idea for the experiment design. This pairwise comparison follows the same method as Idea Arena, refer to Sec. 3.4 for details.

### 2.4 Experiment Design

While our primary goal is to generate novel ideas, it is also useful to develop experimental plans that help users implement these ideas. Thus, we extended the CoI agent to include experiment design. As shown in the lower-right of Figure 2, we prompt the LLM with experiments from existing works obtained from Sec. 2.2 as few-shot examples, along with the proposed idea and key entities, to guide the LLM in designing experiments for our ideas (Table 17).

We also employ a review agent to assess the candidate experiment designs. Its main role is to evaluate the clarity and comprehensiveness of the protocol, ensuring all key elements—such as datasets and models—are clearly specified. Additionally, it checks if the design provides enough detail for practical implementation (Table 18). The review agent provides critical feedback on these aspects, subsequently utilizing this information to conduct further searches for relevant literature (Table 19) to help the LLM refine and enhance its previous experiment design (Table 20). Through this iterative process of review and refinement, we arrive at a final experiment design.

## 3 Experimental Setups

### 3.1 Implementations

In our CoI agent, we primarily use GPT-4o (05-13)[4] as our LLM implementation. For some modules that require full-paper understanding, we use GPT-4o-mini (07-18) to read the paper and summarize the core contents due to its lower price and good summarization capability. We use Semantic Scholar as our academic search engine. For the main experimental results, the maximum length of the CoI is set to 5 and the number of CoI branches is set to 3, and their analysis results are given later. The iteration number of self-refinement in the experiment design stage is set to 1 for cost saving.

### 3.2 Data

To evaluate our CoI agent's ability to generate novel ideas, we collect recent research topics from Hugging Face's Daily Papers[5], known for its timely updates on AI research and the high quality of the featured papers. We select papers submitted between August 1 and September 15, 2024, ensuring that the topics are sufficiently new and the time frame is after the data cutoff of the LLM. We ask 10 skilled researchers with diverse interests in AI to identify papers that capture their interests. Subsequently, we prompt GPT-4o to extract research topics, proposed ideas, and their corresponding experiment designs from these selected papers (Tables 21, Table 22 and 23). The extracted topics will then be returned to the researchers for validation, ensuring that the extracted topics are valid and reasonable within their research domains. The extracted ideas and experiment designs will be utilized as our Real Paper baseline, as described in Section 3.3. Due to the substantial costs associated with generating and evaluating ideas and experiment designs, we adhere to the assessment scale of Lu et al. (2024); Wang et al. (2023) to collect 50 research topics in total for evaluation.

[4] https://openai.com/api/
[5] https://huggingface.co/papers

### 3.3 BASELINES

We compare our CoI agent with recent works on idea generation and experiment design. To ensure a fair comparison, we employ GPT-4o and Semantic Search as the LLM and academic retriever implementations, respectively, across all baseline methods. Furthermore, we unify the output format of the generated ideas and experiment designs to minimize evaluation preference towards more structured outputs (Chiang et al., 2024). We compare with the following baselines:

- **RAG**: This is a vanilla retrieval augmented generation approach (Lewis et al., 2020), where we directly prompt the LLM with retrieved literature for idea generation and experiment design.
- **ResearchAgent** (Baek et al., 2024): This work leverages additional academic knowledge graph for enhancing the literature retrieval and adopts a multi-agent framework to iteratively refine ideas through peer discussions. We follow the original paper to reproduce this baseline.
- **GPT-Researcher** (Assafelovic, 2023): GPT-Researcher is an agent framework specifically designed for the research domain. The agent is enhanced with plan-and-solve and RAG capabilities.
- **AI-Scientist** (Lu et al., 2024): This work originally aims to generate the entire paper with the idea, methods, and experimental results. We extract the components related to idea generation and experiment design to serve as our baseline.
- **Real Paper**: Note that, in Sec. 3.2, we extract topics from existing research papers. Therefore, the ideas and the experiment designs from these papers serve as a natural baseline to quantify the gap between model-generated ideas and genuine human ideas.

### 3.4 EVALUATION: IDEA ARENA

**Model-based Evaluation.** The open-ended nature of idea generation poses challenges for automatic evaluation. Prior work primarily uses LLM-based Likert scale system to score ideas (Baek et al., 2024; Lu et al., 2024). However, Si et al. (2024) show this method poorly aligns with human preferences. Instead, they show LLMs perform better in ranking ideas. To obtain reliable scores for evaluation, we propose Idea Arena, a pairwise evaluation system using a Round-Robin tournament to compute ELO scores for each idea-generation method. For a given topic, we require the LLM judge to rank the ideas generated by any pair of methods (Table 24). We evaluate each pair twice with order reversed to reduce the position bias. To comprehensively evaluate an idea from multiple perspectives, we incorporate criteria from ICML 2020 review guidelines [6], and those in Si et al. (2024), which consist of Novelty, Significance, Clarity, Feasibility, and Expected Effectiveness. Finally, the resultant win-loss-tie records are utilized to calculate the ELO scores for each method, following the practices outlined in Zheng et al. (2024); Zhao et al. (2024). We also evaluate the experiment design in the same pairwise way, focusing on Feasibility, Technical Quality, and Clarity. Refer to Definitions for all metrics in Tables 5 and 6 of the Appendix.

**Human Evaluation.** We also perform human evaluation in a pairwise manner. The 10 researchers who review the extracted topics are asked to rank two ideas and experiment designs using the same criteria. To ensure fairness, we anonymize the source of the ideas by concealing the method identity.

## 4 RESULTS

### 4.1 IDEA GENERATION

**Main results.** Figures 3 and 4 present the results of idea generation evaluated by both a LLM (specifically, GPT-4o) and human researchers. Detailed scores are in Table 26 of Appendix. Overall, our CoI agent demonstrates superior performance compared to all other automated methods in both model-based and human-based evaluations. Notably, It substantially outperforms the second-best baselines, GPT-Researcher and RAG, by margins of 108 and 56 ELO scores, respectively, in the two evaluation settings. Our CoI agent's performance is on par with that of the Real Paper baseline and even excels in the metrics of Novelty and Significance. These results highlight its exceptional capabilities in idea generation. Furthermore, CoI demonstrates superior performance in Clarity, Feasibility, and Expected Effectiveness compared to other automated methods in human evaluation. Nevertheless, it still lags considerably behind the Real Paper in these areas. This substantial gap

[6] https://icml.cc/Conferences/2020/ReviewerGuidelines